

PONTA DELGADA AP, Portugal

WMO: 085120

Lat: 37.7442N

Lon: 25.6969W

Elev: 255

StdP: 14.56

Time Zone: -1.00 (AZR)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	48.6	50.4	39.5	35.9	52.5	41.1	38.4	53.4	33.8	58.5	31.1	58.1	7.2	0	0.425

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.3	77.5	70.8	76.8	70.4	75.5	69.5	73.3	75.6	72.5	74.9	71.6	74.1	9.6	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.3	120.7	74.6	71.6	117.8	74.1	70.4	112.8	73.2	36.9	76.1	36.2	74.4	35.4	74.4	79.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 28.3	24.6	21.4	DB	45.0	80.5	1.6	1.3	43.8	81.4	42.8	82.2	41.9	82.9	40.7	83.8	(4)
(5)			WB	42.1	75.2	1.6	1.6	40.9	76.4	39.9	77.4	39.0	78.3	37.9	79.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.6	58.7	57.8	58.0	59.2	61.9	65.8	69.8	72.0	70.5	66.7	62.3	60.1	(6)	
	DBStd	5.65	2.89	2.70	2.70	2.57	2.39	2.45	2.31	2.00	2.39	3.05	3.23	3.14	(7)	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD65	1166	195	201	217	175	98	19	0	0	0	17	91	153	(9)	
	CDD50	4970	270	219	248	275	369	475	615	682	616	518	370	313	(10)	
	CDD65	662	0	0	0	0	3	44	150	217	166	70	11	1	(11)	
	CDH74	680	0	0	0	0	0	4	116	371	175	13	1	0	(12)	
	CDH80	2	0	0	0	0	0	0	0	2	0	0	0	0	(13)	
(14) Wind	WSAvg	10.2	11.4	11.3	12.2	11.2	9.7	9.0	7.8	8.0	8.5	10.7	10.3	12.0	(14)	
(15) Precipitation	PrecAvg	40.4	5.1	4.1	3.8	2.8	2.2	1.5	1.1	1.7	3.3	4.5	5.1	5.4	(15)	
	PrecMax	57.4	11.3	9.8	7.9	7.9	7.6	4.4	4.3	5.6	9.3	12.0	11.1	17.9	(16)	
	PrecMin	30.5	0.7	0.9	0.2	0.4	0.1	0.2	0.0	0.2	0.5	0.8	0.4	0.7	(17)	
	PrecStd	7.1	2.4	2.2	1.9	1.9	1.7	0.9	0.8	1.3	2.0	2.7	2.5	3.1	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	65.4	64.7	64.8	67.7	69.9	74.2	78.5	79.7	78.6	75.5	71.1	68.0	(19)	
		MCWB	62.2	60.5	60.0	60.9	64.5	67.6	70.7	71.5	71.9	70.4	66.1	64.4	(20)	
	2%	DB	64.5	63.5	63.9	65.7	68.2	72.6	76.6	78.4	77.1	73.7	69.3	66.4	(21)	
		MCWB	61.5	59.9	59.7	60.8	63.2	66.7	69.8	71.2	71.1	68.7	65.0	63.3	(22)	
	5%	DB	63.6	62.7	62.8	64.4	67.1	71.4	75.3	77.1	75.7	72.6	68.1	65.8	(23)	
		MCWB	60.8	59.5	59.1	60.0	62.4	66.0	69.0	70.5	70.0	68.0	64.4	62.8	(24)	
	10%	DB	62.8	62.0	62.2	62.9	66.1	70.0	73.8	75.9	75.0	71.6	66.9	64.6	(25)	
		MCWB	60.3	58.9	58.7	59.2	61.7	65.0	68.1	69.7	69.5	67.4	63.4	61.8	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	63.6	62.5	62.4	63.6	66.4	70.4	73.2	74.7	74.5	72.4	68.0	65.6	(27)	
		MCDB	64.5	63.5	63.6	65.4	68.0	72.4	75.8	76.9	76.7	74.2	69.4	66.9	(28)	
	2%	WB	62.6	61.6	61.5	62.1	65.1	68.9	72.2	73.6	73.0	70.8	66.9	64.5	(29)	
		MCDB	63.6	62.7	62.8	64.0	66.9	70.9	74.6	76.0	75.2	72.7	68.4	65.8	(30)	
	5%	WB	61.9	60.7	60.4	61.5	64.0	67.6	71.1	72.7	72.2	69.3	65.6	63.6	(31)	
		MCDB	63.1	62.0	61.9	63.3	66.0	69.7	73.5	75.3	74.6	71.5	67.4	65.0	(32)	
	10%	WB	61.2	59.9	59.8	60.5	63.3	66.7	70.2	71.8	71.1	68.6	64.6	62.7	(33)	
		MCDB	62.6	61.5	61.5	62.4	65.4	69.0	72.7	74.6	73.7	70.8	66.7	64.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	6.5	7.1	7.3	7.7	8.0	8.3	8.9	9.3	8.8	7.8	7.3	6.7	(35)	
		MCDBR	5.9	7.0	7.3	8.2	8.7	9.2	9.5	9.9	9.3	8.2	7.5	6.2	(36)	
	5% WB	MCWBR	5.7	6.2	6.3	6.4	6.2	6.5	6.1	6.3	6.3	6.2	6.4	5.9	(37)	
		MCWBR	5.0	5.8	6.1	6.7	6.9	7.7	7.8	8.0	7.6	7.0	6.8	5.7	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.347	0.353	0.375	0.377	0.370	0.372	0.366	0.362	0.361	0.368	0.351	0.352	(40)	
		taud	2.532	2.517	2.453	2.455	2.502	2.530	2.556	2.593	2.599	2.557	2.549	2.510	(41)	
	Ebn at Noon	Ebn at Noon	265	279	282	286	289	287	288	288	283	270	261	254	(42)	
		Edh at Noon	25	29	33	35	34	33	32	30	28	27	24	24	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	675	947	1306	1655	1948	2046	2063	1947	1564	1091	757	581	(44)	
	RadStd	RadStd	52	58	81	127	116	121	100	65	74	60	47	39	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	-0.56	N/A	N/A	N/A	N/A	N/A	N/A
Regional (3 neighbors)	N/A	N/A	N/A	N/A	+0.52	+0.77	N/A	N/A	N/A	N/A

Nomenclature: See separate page